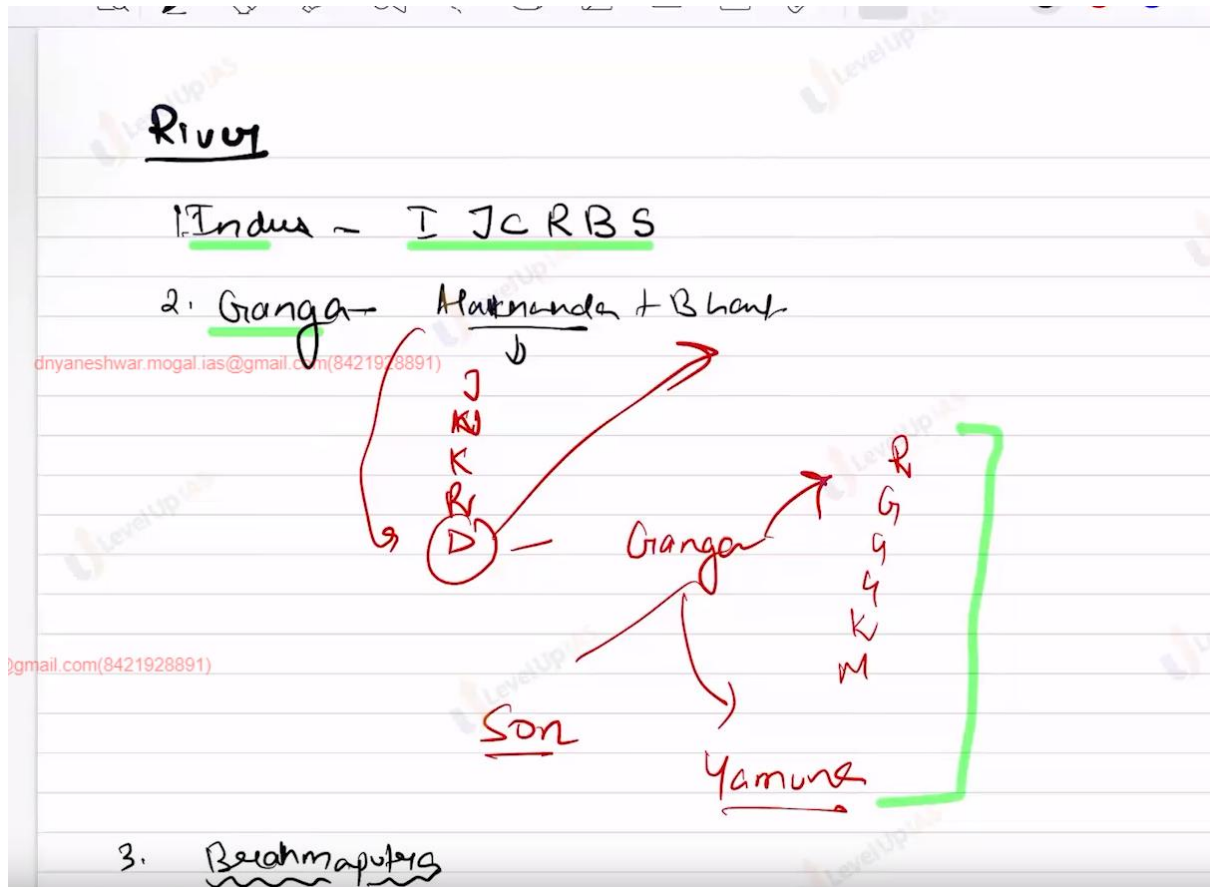


(31) Rivers-II



Mapping India:

1. Tides refers to vertical movement of water nearly each place on the earth experiences two high tide tow low tide

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#2) River Son

- Origin: Amarkanthak Plateau
- Tributary: Rihand and North Koel
- Dam: Rihand Dam (aka Ballabh Pant Sagar Dam) on Rihand River in UP;
- Bansagar Dam on River Son in U.P



The map shows the Son River (labeled 'Son River') flowing through central India. It is a tributary of the Ganga River. Other rivers shown include the Yamuna River, Chambal River, and Ken River. The word 'INDIA' is written in large letters at the bottom of the map. A green dot marks the location of the Son River's origin in the Amarkanthak Plateau region.

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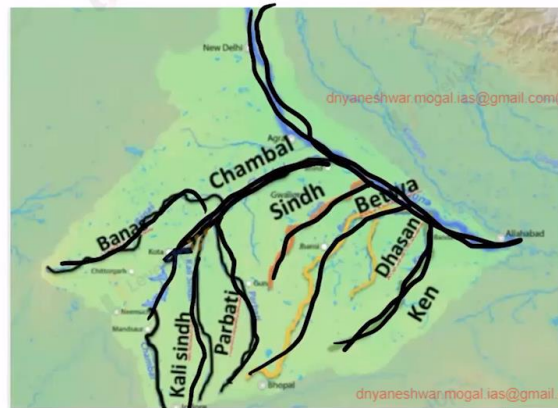
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- Bansagar Dam on River Son in U.P



Chambal River

- Vindhayan Range
- National Chambal Sanctaury is at confluence of Chambal and Yamuna.
- It is famous for Gharial, Red Crowned Turtle
- Chambal forms the boundary between Rajasthan & M.P ; and also between M.P & U.P.
- Kota is on Chambal River



Gharial:

Does not attack humans

Feed on corpses

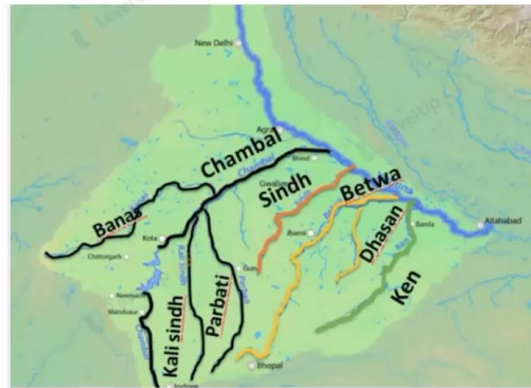
Gavial or fish-eating crocodile





Chambal River

- Tributaries:
 - Left: Banas (Origin: Aravalli Range)
 - Right: Kalisindh (Origin: Vindhayan Range), Parbati (Origin: Vindhayan Range), Kshipra (Origin: Vindhayan Range near Ujjain, joins Chambal at M.P Raj Border, Famous for Kumbh)
- Chambal & Banas meets at Ranthambore National Park in Rajasthan
- Gandhi Sagar Dam is on Chambal River (MP)



Betwa.

- Origin: Vindhyan Range
- Dam: Parichha
- Tributary: Dhasan River rises from Vindhyas in M.P

Ken

- Origin: Vindhyas near Banher Range
- Falls: Raneh Falls
- It has Ken Gharial Sanctuary and Panna National Park
- Ken Betwa Link: Dhaudhan Dam: River from Ken to Betwa



Practise Question:

Q) Dhauli meets alakananda at?

Joshimath / Vishno math

Q) Mandakini meets at? - *B R*

Q) Tehri Dam is on? - *B*

Q) Rajaji NP is on? *Ganga*

Q) Origin of Ghagra? *✓*

Q) Origin of Son? *✓*

Q) Chambal & Banas meets at Ranthambore National Park in Rajasthan

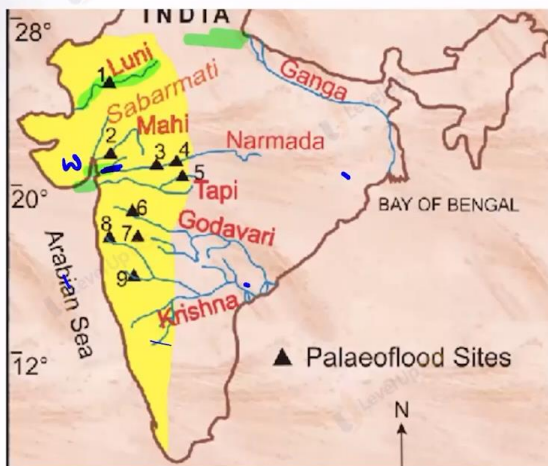
Q) Ken Betwa Link. Which Dam

The superlative

Let's practise!

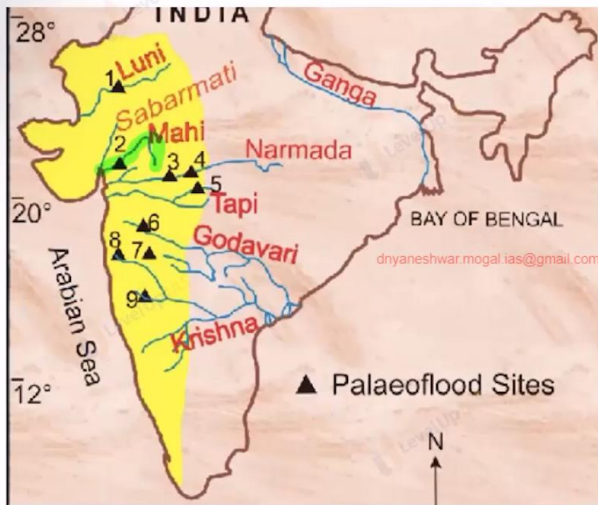


Some Small Rivers



1) Luni River

- **Origin:** Aravalli Range near Pushkar Lake in Ajmer Rajasthan
- Originate in 2 branches: Saraswati & Sabarmati which join at Govindgarh
- Lake: Sambhar Lake, Pushkar Lake and Pachpadra Lake
- Passes via Thar Desert and end in Rann of Kutch
- It is the largest river system of Rajasthan



#3) Mahi River

- Origin: Bagh Hills in MP
- Dam: Mahisagar Dam in Rajasthan
- Mahi encircles Ratan Mahal Wildlife Sanctuary in Gujarat. It is a sloth bear sanctuary
- It cuts tropic of cancer twice
- Drains in Gulf of Khambat

Narmada:

- Origin: Amarkanthak Plateau
- Passes via M.P, Mah & Gujarat
- Flows via Rift Valley
- Between Satpura and Vindhya and drains in Gulf of Khambat
- Tributaries: Tawa (Rises from Betul)
- Waterfall: Dhuandhar Waterfall; Marble Rock



Narmada:

- Origin: Amarkanthak Plateau
- Passes via M.P, Mah & Gujarat
- Flows via Rift Valley
- Between Satpura and Vindhya and drains in Gulf of Khambhat
- Tributaries: Tawa (Rises from Betul)
- Waterfall: Dhuandhar Waterfall; Marble Rock

Jabalpur MP

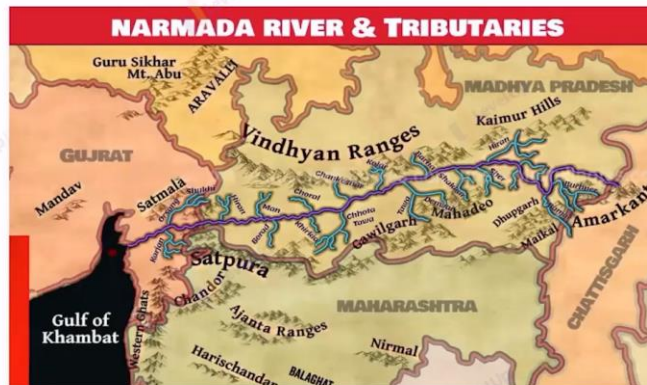
MP

Gorge



Narmada:

- Longest West Flowing River
- Aliabet is estuary of Narmada formed at the mouth of River
- Dam:
 - Sardar Sarovar Dam in Gujarat
 - Indira Sagar Dam in MP
 - Gandhi Sagar Dam in MP
 - Omkareshwar Dam in M.P
 - Tawa Reservoir in MP
- Narmada passes via Mandla Fossil National Park in M.P



- origid

- E to W

- ~~dia~~ - ai Estuary - H126

- Dam - 
MP

- Gulf of Phambet

- Tawa

MP

- Gulf of Khambhat
- Tawa
- Panna NP / Mandla
- Jabalpur - Phaw
- Marble Rocks

2013

The Narmada river flows to the west, while most other large peninsular rivers flow to the east. Why?

1. It occupies a linear rift valley.
2. It flows between the Vindhyas and the Satpuras.
3. The land slopes to the west from Central India.

Select the correct answer using the codes given below.

- (a) 1 only
- (b) 2 and 3
- (c) 1 and 3
- (d) None

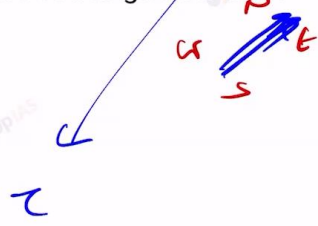
2013

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1. It occupies a linear rift valley. ✓
2. It flows between the Vindhyas and the Satpuras. ✓
3. ~~The land slopes to the west from Central India~~

Select the correct answer using the codes given below.

- (a) 1 only
- (b) 2 and 3
- (c) 1 and 3
- (d) None



Tapi

- Origin: Betul district in MP
- Passes via MP, Mah and Gujarat
- Tributaries: Purna, Girna
- Dam: Ukai Dam
- WLS: Yawal WLS (Mah),
- Melghat Tiger Reserve (MP-MAH border)



West flow

1. short river .
2. west coast - high elevation $\therefore$ no deposition by river
Tide
3. waves are powerful sweep sediments

East flow

- long river $\therefore$ sediment
- Elevation is less $\therefore$ more deposition
- Tide
3 waves allow deposition

> Godavari

- Origin: Trimbakeshwar in Nashik district in Maharashtra
- Main River and Tributaries flows through Maharashtra, Telangana, Andhra, Chhattisgarh, Odisha
- Largest Peninsular River: aka Dakshin Ganga



Godavari

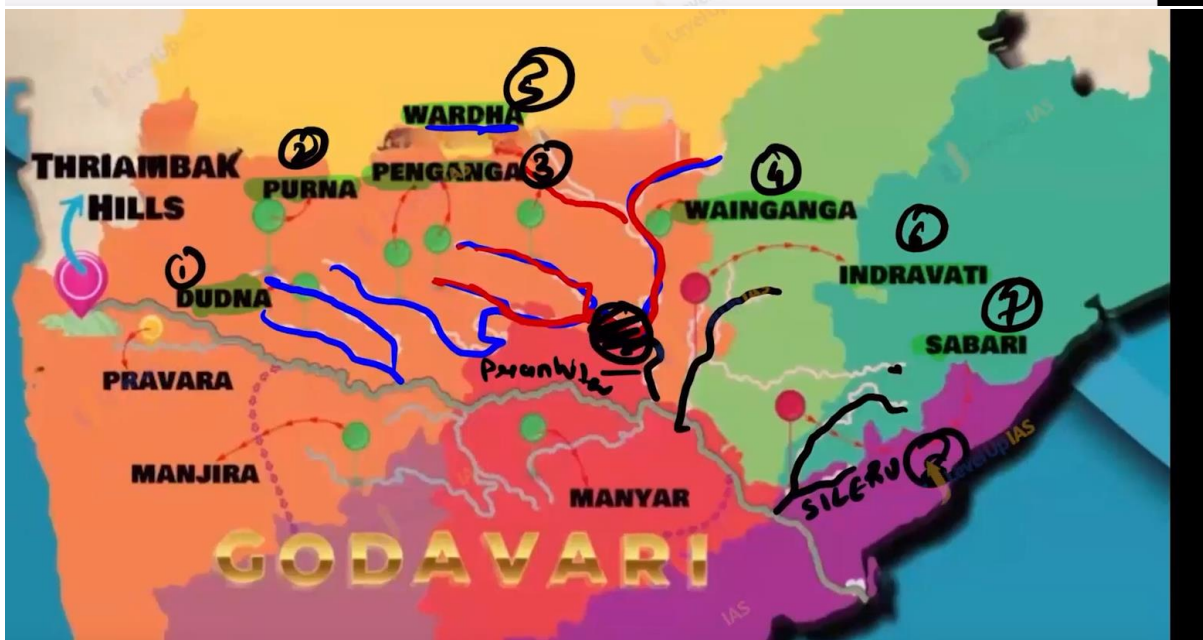
- Left bank tributaries:

- Left: Dudhna, Purna, Peinganaga Weinganga, Indravati (Indravati NP), Sabari, Silleru, Pranhita (Kaleshwaram lift irrigation system)

- Right bank tributaries: Manjhra, Pravara



Pranhita River is the largest tributary of Godavari River covering about 34% of its drainage basin conveying the combined waters of the Penganga River, the Wardha River, and the Wainganga River



- Pelingang
- Weingang

Majra
Pranava

(12)

- Purna
- Dodha
- Wandha
- Pranahita
- Cabaru
- Sillu

Manya

- Indravati

2015: UPSC

Consider the following rivers :

1. Vamsadhara ✕
2. Indravati
3. Pranahita
4. Pennar

Which of the above are tributaries of Godavari ?

- (a) 1, 2 and 3
- (b) 2, 3 and 4
- (c) 1, 2 and 4
- (d) 2 and 3 only

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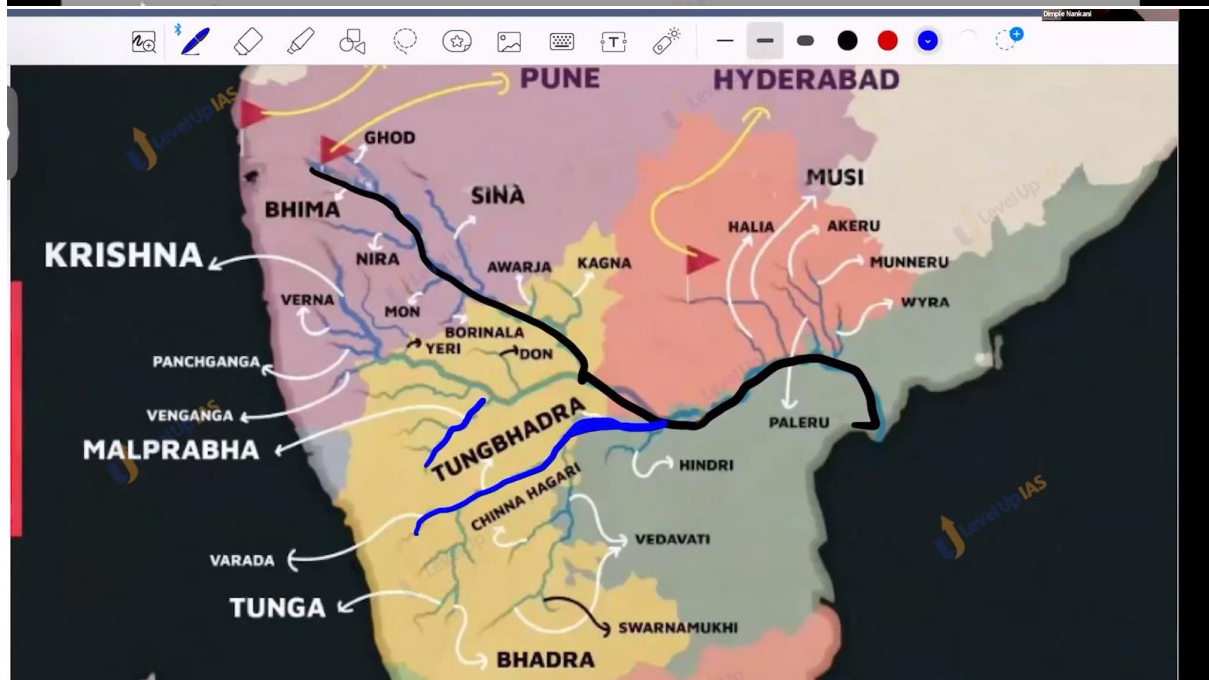
Krishna River

- Origin: Mahabaleshwar in Maharashtra
- Right bank: Ghat-prabha, Tungabhadra, Mallaprabha.
- Left: Bhima, Musi
- River Koyna rises in Mahabaleshwar and Joins Krishna
- Bhima river is also known as **ChandraBhaga** river.
- Bhima is the **longest** (largest is tungbhadra) tributary of the river Krishna.



KRISHNA RIVER MAP

The map shows the Krishna River originating in the Western Ghats near Nashik, Maharashtra, and flowing eastwards. Major tributaries on the left bank include the Bhima, Musi, and Tungabhadra. On the right bank, tributaries like the Ghatprabha and Mallaprabha are shown. The river eventually empties into the Bay of Bengal. Key cities like Mumbai, Pune, and Hyderabad are marked. A watermark 'Dnyaneshwari@gmail.com' is visible across the map.



Krishna River

- Origin: Mahabaleshwar in Maharashtra
- Right bank: Ghat-prabha, Tungabhadra, Mallaprabha.
- Left: Bhima, Musi
- River Koyna rises in Mahabaleshwar and joins Krishna
- Bhima river is also known as ChandraBhaga river. - Pandaripani
- Bhima is the **longest** (largest is tungbhadra) tributary of the river Krishna.



#7) Krishna River

- Krishna has: Almati dam, Srisalam dam, Nagarjuna sagar dam are on Krishna River
- It has Nagarjuna Srisagar Tig Reseve and Srisalem Tiger Reserve
- Musi river has Hyderabad city
- Vijaywada is on Krishna river



#7) Krishna River

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- It has Nagarjuna Srisagar Tiger Reseve and Srisalem Tiger Reserve
- Musi river has Hyderabad city
- Vijaywada is on Krishna river



Places: 2019 - Last

Famous place: River

1. Pandharpur : Chandrabhaga
2. Tiruchirapalli : Cauvery
3. Hampi : Malaprabha

Which of the pair given above are correctly matched

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Places: 2019 - Last

Famous place: River

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Level Up Class

Places: 2019 - Last

Famous place: River

1. Pandharpur : Chandrabhaga, Mahe

2. Tiruchirapalli : Cauvery, TN

3. Hampi : ~~Malaprabha~~ Tungabhadra, Krishna

Which of the pair given above are correctly matched

(a) ~~1 and 2 only~~

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Level Up Class

Answer: a

The Chandrabhaga (Bhima is also known as Chandrabhaga) river flows through the Pandharpur. So #1 is right.

Tiruchirappalli, situated on the banks of the river Cauvery is the fourth largest city in Tamil Nadu. So #2 is right.

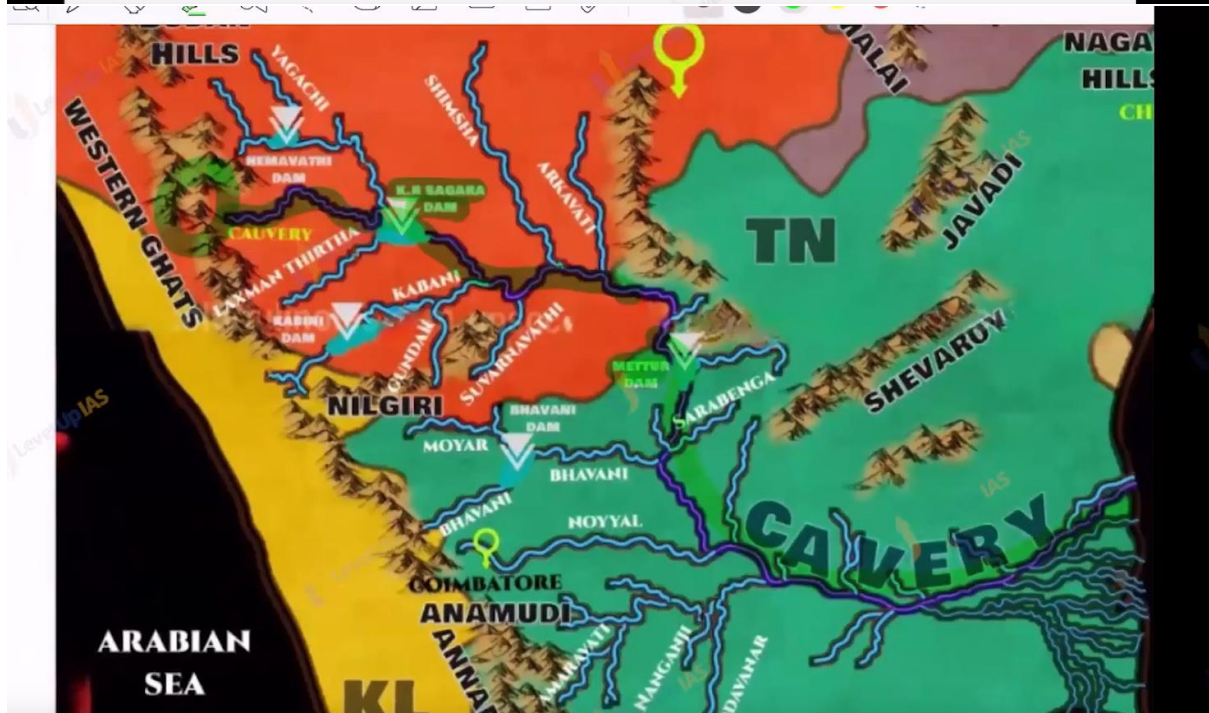
Hampi, the 14th century capital of one of the greatest empires of medieval India called the Vijayanagar Empire, lies in the state of Karnataka protected by the tempestuous river Tungabhadra. So #3 is wrong

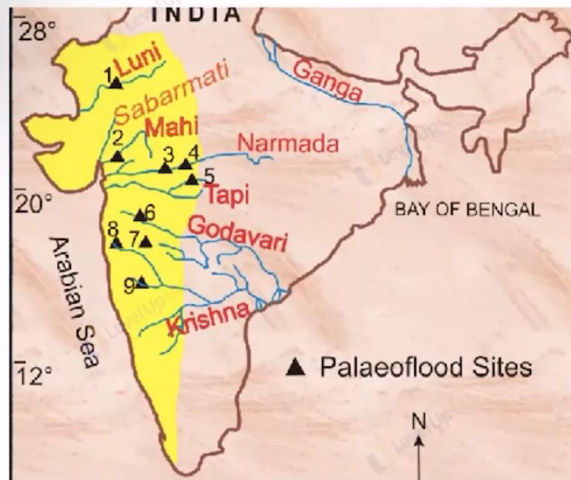


Level Up Class

Kaveri River

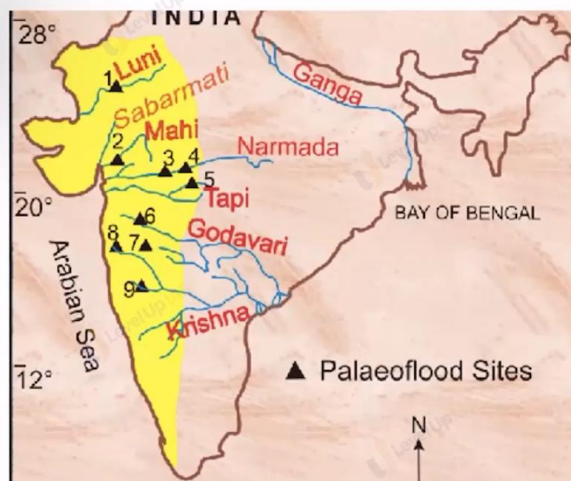
- Origin: Tala Kaveri in Brahmagiri Hills
- Tributary: Amravati, Hemavati, Kabini, Shimsha, Bhavani, Noyil
- Dam: Krishnaraj sagar dam, Mettur dam, bhavanisagr dam on bhavani river
- Falls: Shivasamudram falls
- Hongenekkal Falls
- Srirangam Island and Shivasamudram island



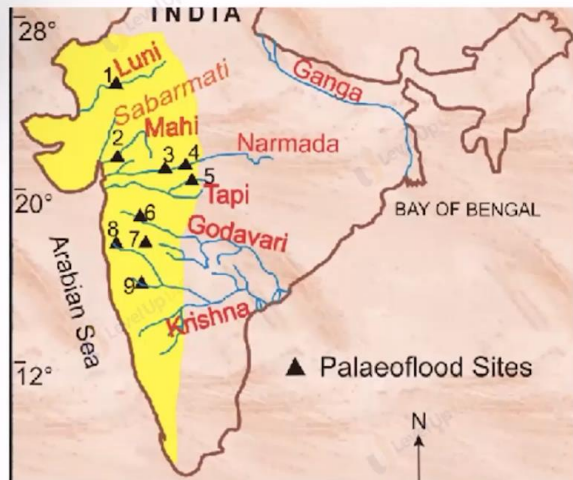
) Vagai

- Origin: Varusunadu hills
- Dam: Vaigai Dam
- Empties into Palk Strait



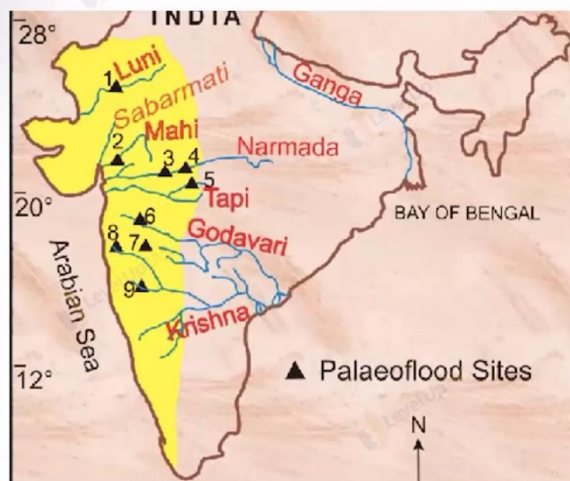
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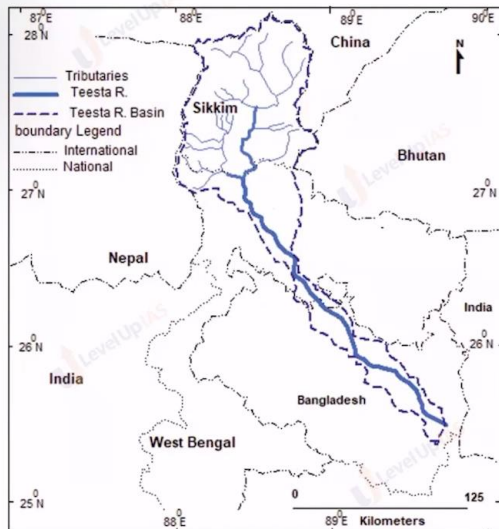
) Vagai

- Origin: Varusunadu hills
- Dam: Vaigai Dam
- Empties into Palk Strait



#) Periyar Reserve

- Longest river of Kerala (ORIGIN: Kerala and flows through Kerala only)
- It is between Palani Hills and Cardamom Hills
- Dam: Idukki Dam, Mullaperiyar dam
- Lake: Vembanad Lake, Vembanad wetland
- Sabarimala temple is in Periyar Tiger Reserve



#) Teesta

- Origin: Tso-lhamo lake, cholamu lake, Zemu Glacier
- Tributary of Brahmaputra
Rangit river is tributary of Teesta
- Dam: Teesta Barrage
- BR: Kunchenjunga BR
- Flows through sikkim and West Bengal only and meets Brahmaputra in Bangladesh

2017: River Teesta

With reference to river Teesta, consider the following statements

1. The source of river Teesta is the same as that of Brahmaputra but it flows through Sikkim
2. River Rangeet originates in Sikkim and it is a tributary of river Teesta.
3. River Teesta flows into Bay of Bengal on the border of India and Bangladesh.

Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 2 only**
- (c) 1 and 3 only
- (d) 1, 2 and 3

UP IAS Level Up Class

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UP IAS Level Up Class

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(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

#) Damodar

- Origin: CNP
- Divide CNP in two parts: Hazaribagh and Ranchi
- Tributaries: Barakar, konar
- Dam: Tillaya and Maithon; Barakar, Panchet; Damodar, Konar; Konar River
- DVC - Damodar Valley Corporation

#) Subarnarekha

- Origin: Ranchi along CNP
- Falls: Hundru Falls



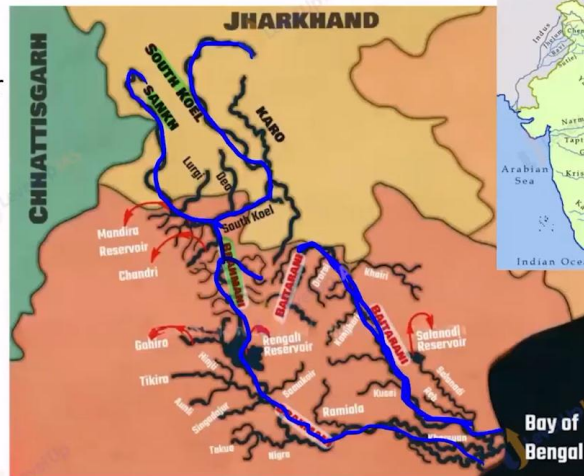
CNP - Damodar
- Subarnarekha
- Barahmani
- Baitarni
- Mahanadi

#) Baitarni

- Gonasika hills of Keonjhar district
- Origin: Garjhat Hills
- Passes through Bhitarkanika

#) Brahmani

- Brahmani river is formed by the combined waters of Koel and Sankh rivers which originate in CNP
- Passes through Bhitarkanika
- It has wheeler island (Abdul Kalam Island)



#) Baitarni

- Gonasika hills of Keonjhar district
- Origin: Garjhat Hills
- Passes through Bhitarkanika NP - Salt water lagoon

#) Brahmani

- Brahmani river is formed by the combined waters of Koel and Sankh rivers which originate in CNP
- Passes through Bhitarkanika
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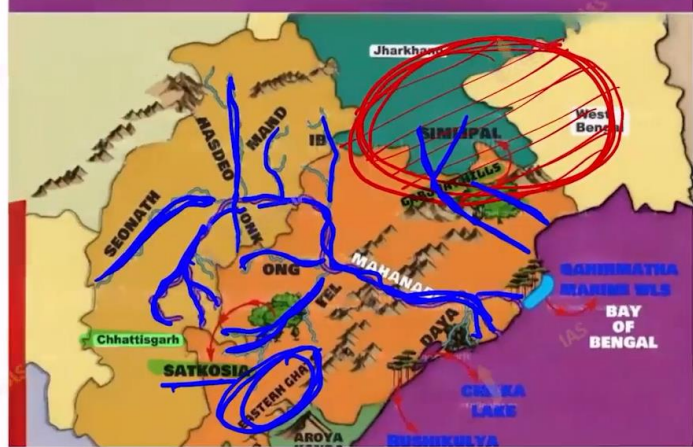
↳ Bhitarkanika to sea

#) Mahanadi

- Origin: Dandakarnya plateau
- Tributaries: Hasdeo, Seonath, Mand, Ib, Tel
- Flows via Satkosia Tiger Reserve in Odisha
- Flows via Chattisgarh and Odisha
- Hirakud Reservoir



MAHANADI RIVER MAP



2021

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Consider the following Rivers:

1. ~~Brahmani~~
2. Nagavali
3. ~~Subarnarekha~~
4. Vamsadhara

Which of the above rise from the Eastern Ghats?

- (a) 1 and 2
- (b) 2 and 4
- (c) 3 and 4
- (d) 1 and 3

2021

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Consider the following Rivers:

1. ~~Brahmani~~
2. ~~Nagavali~~
3. ~~Subarnarekha~~
4. ~~Vamsadhara~~

Which of the above rise from the Eastern Ghats?

- (a) 1 and 2
- (b) 2 and 4
- (c) 3 and 4
- (d) 1 and 3

Quick Revision

-----: Hemis High Altitude National

-----: Has wular lake, Dal Lake, Srinagar

-----: Made of Chandra and Bhaga: Both meets at Keylong: Hence called as Lady of Keylong

Beas and Sutlej meet at ----- wetland

----- **Origin:** It passes via Valley of Flowers

Ganga Passes via ----- National Park

----- : It passes via Corbet National Park

National Chambal Sanctuary is at confluence of ----- and -----

Chambal & Banas meets at ----- National Park in Rajasthan

2019: Rivers

Consider the following Pairs :

Glacier: River

1. Bandarpunch : Yamuna
2. Bara Shigri : Chenab
3. Milam : Mandakini
4. Siachen : Nubra
5. Zemu : Manas

Which of the following pairs given above are correctly matched?

- (a) 1,2 and 4
- (b) 1,3 and 4
- (c) 2 and 5
- (d) 3 and 5

2019: Rivers

1. Siachen Glacier, one of the world's longest mountain glaciers, lying in the Karakoram Range system of Kashmir. It is the source for Nubra River
2. Bandarpunch: Yamuna as correct.
3. River Mandakini actually originates from the springs fed by melting snow of Chorabari glacier about one km above Kedarnath temple.
4. Ncert: Sarda or Saryu river rises in the Milam glacier in the Nepal Himalayas
5. Bara Shigri is the largest glacier located in Lahaul Spiti region in Chandra Valley, Himachal Pradesh. It is a 30-km long glacier, the second longest glacier in the Himalayas after Gangotri. It flows northwards and feeds the Chenab river.

2022

Gandikota canyon of South India was created by which one of the following rivers ?

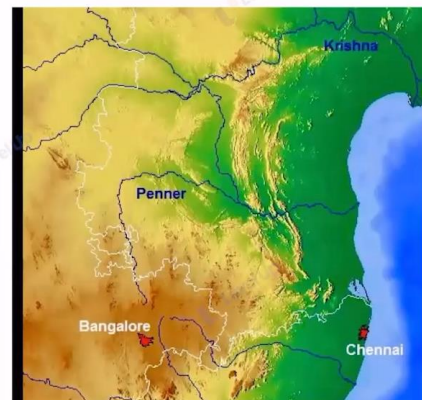
- (a) Cauvery
- (b) Manjira
- (c) Pennar
- (d) Tungabhadra

CASE STUDY



Figure 4.10 The Canyon of Gandikota, the Pennar River in Andhra Pradesh

Gandikota, Kadapa district of Andhra Pradesh is known for its spectacular gorge formed by river Pennar that cuts through the Erramala hills. This handsome piece of Nature's architecture is known as the **Hidden Grand Canyon of India**. Magnificent Gandikota fort is located majestically on top of this gorge. Belum Cave found here is the second largest cave system in India. In fact, geologists have also found **surplus deposits of Quartz** in the stalactite and stalagmite formations of the cave. Adjacent to Gandikota fort, lies a magnificent lake that is believed to have been established by emperor Sri Krishnadevaraya using water from the Pennar river.



CASE STUDY

TN Text Book



Figure 4.10 The Canyon of Gandikota, the Pennar River in Andhra Pradesh

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2022

Consider the following pairs:

Reservoirs : States

1. Ghataprabha : Telangana
2. Gandhi Sagar : Madhya Pradesh
3. Indira Sagar : Andhra Pradesh
4. Maithon : Chhattisgarh

How many pairs given above are not correctly matched?

- (a) Only one pair
- (b) Only two pairs
- (c) Only three pairs
- (d) All four pairs

2022

Consider the following pairs:

Reservoirs : States

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2022

Consider the following pairs:

Reservoirs : States

1. ~~Ghataprabha~~ : ~~Telangana~~
2. ~~Gandhi Sagar~~ : ~~Madhya Pradesh~~
3. ~~Indira Sagar~~ : ~~Andhra Pradesh~~
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How many pairs given above are not correctly matched?

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Mains Questions: 2013-2022

2013: There is no formation of deltas by rivers of the Western Ghats. Why?

2013: Bring out the causes for the more frequent occurrence of landslides in the Himalayas than in the Western Ghats

2014: Bring out the relationship between the **shrinking Himalayan glaciers and the symptoms of climate change** in the Indian sub-continent.

2016: "The **Himalayas are highly prone to landslides.**" Discuss the causes and suggest suitable measures of mitigation.

2021: Why is India considered as a sub-continent? Elaborate your answer. (Answer in 150 words) 10

2022: Discuss the natural resource potentials of 'Deccan trap'. (Answer in 150 words) 10

